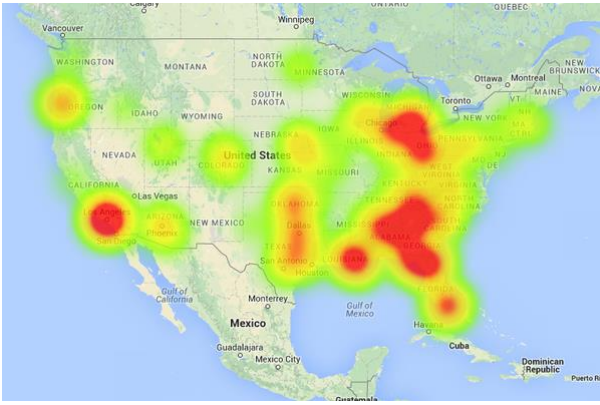


6 Types of Thematic Maps:

These are maps that use ONE characteristic to create a THEME on a map.



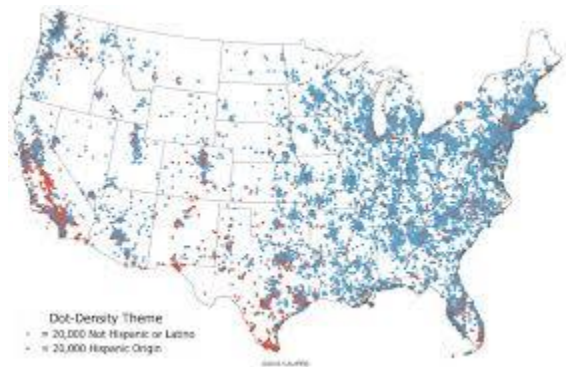
Heat Map: a representation of data in the form of a map or diagram in which data values are represented as colors by intensity of the observations



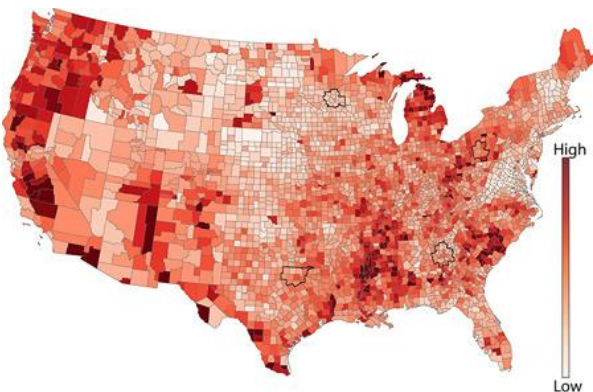
Graduated Symbol Map: uses some symbol to display the frequency of an observation. The larger the symbol on the map, the higher the frequency of that observation found in the geographic unit (county, state, etc.)



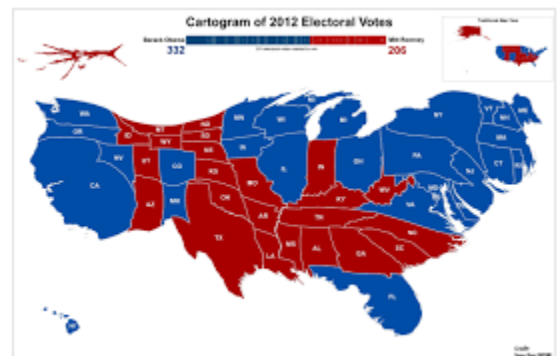
Isoline Map: displaying lines that connect points of equal value. Every observation along the same line will have the same value.



Dot Density Map: uses dots to represent the frequency of an occurrence. Each dot will represent a set number of occurrences.



Choropleth Map: shows a pattern of a occurrences in a geographic unit (county, state, etc.), such as population density or voting patterns, by using various colors or degrees of shading.



Cartogram Map: uses proportionality (i.e., area on the map) to show intensity of a particular observation. The size of the geographic unit is adjusted, which warps the shape of the study area.

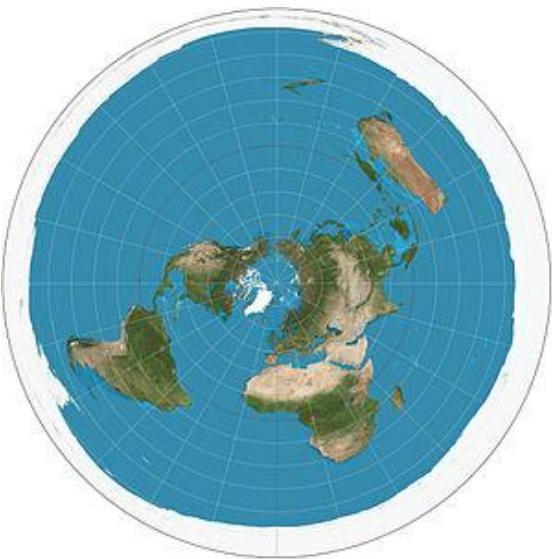
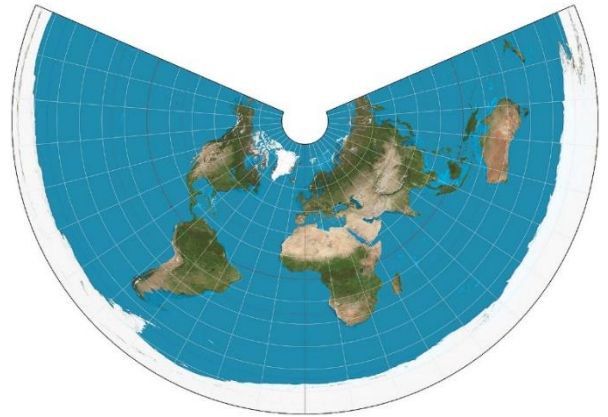
Map Projections:

How a 3D sphere is represented over a 2D piece of paper! They each have their own distortions, but also have situations where they are best used.



Mercator/Cylindrical: a map projection that fairly accurately shows shape and direction, but distorts distance and size of land masses. This map is most accurate at the equator, but the area around the poles are greatly exaggerated (In reality, Greenland is about the same size as Mexico)

Conic: a map projection of the globe onto a cone with its point over one of the earth's poles. It is most accurate in the mid-latitudes



Polar/Azimuthal: a map which shows true compass directions; longitude lines are straight and latitude lines are circles. The map is most accurate at the pole and distorts shape and size more toward the outer edges. (Azimuthal means direction and is measured in degrees (like an angle))

Compromise/Robinson: a projection that maintains overall shapes and relative positions without extreme distortion. This is done by **slightly** distorting **every** characteristic in order to limit the overall distortion.



Equal Area Map Projection: The **equal area projection** retains the relative size of the area throughout a map.

So that means at any given region in a map, it keeps the true size of features.

While equal area projections preserve area, it distorts shape and angles.

This is just one example of an equal area projection.

